

Prueba Objetiva 2

$$1) \frac{\sqrt{x^2+1}}{(x^2+3)^2}$$

$$f'(x) = \frac{1}{2\sqrt{x^2+1}} \cdot 2x(x^2+3)^2 - \sqrt{x^2+1} (2(x^2+3) \cdot 2x)$$

$$f'(x) = \frac{1}{\sqrt{x^2+1}} \cdot x(x^2+3)^2 - \sqrt{x^2+1} (4x(x^2+3))$$

$$f'(x) = \frac{x(x^2+3)^2}{\sqrt{x^2+1}} - \sqrt{x^2+1} (4x^3+3)$$

$$f'(x) = \frac{x(x^4+6x^2+9)}{\sqrt{x^2+1}} - 4x^3\sqrt{x^2+1} - 12x\sqrt{x^2+1}$$

$$f'(x) = \frac{x^5+6x^3+9x}{\sqrt{x^2+1}} - 4x^3(x^2+1) - 12x(x^2+1)$$

$$f'(x) = \frac{x^5+6x^3+9x - 4x^5 - 4x^3 - 12x^3 - 12x}{\sqrt{x^2+1}}$$

$$f'(x) = \frac{-3x^5 - 10x^3 - 3x}{\sqrt{x^2+1}} = \frac{-3x^5 - 10x^3 - 3x}{(x^2+3)^4 \sqrt{x^2+1}}$$

* No encontré la respuesta en el portal

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$$2) 3x^2 + 7y^2 = 4$$

$$6x + 14yy' = 0$$

$$14yy' = -6x$$

$$y' = -\frac{6x}{14y} = -\frac{3x}{7y}$$

$$-\frac{3x}{7y} = 1$$

$$\begin{aligned} -3x &= 7y \\ -\frac{3x}{7} &= y \end{aligned}$$

$$\text{recta} = x - 2y + 0 = 0$$

$$m = 1$$

$$3x^2 + 7\left(-\frac{3}{7}x\right)^2 = 4$$

Quadratica =

$$\begin{aligned} x_1 &= -0.966 \\ x_2 &= 0.966 \end{aligned} \quad \pm \frac{\sqrt{270}}{75}$$

Para encontrar y =

$$3\left(\pm \frac{\sqrt{270}}{75}\right)^2 + 7y^2 = 4$$

Quadratica =

$$y = \pm \frac{\sqrt{270}}{35} \approx \pm 0.474$$

$$3) f(x) = (5x^4 + 3x)^2 \sqrt{3x^2 + 2}$$

$$f'(x) = 2(5x^4 + 3x)(20x^3 + 3) \cdot \sqrt{3x^2 + 2} + \frac{1}{2\sqrt{3x^2 + 2}} (6x) \cdot (5x^4 + 3x)^2$$

$$f'(x) = 2\sqrt{3x^2 + 2} (20x^3 + 3)(5x^4 + 3x) + \frac{3x(5x^4 + 3x)^2}{\sqrt{3x^2 + 2}}$$

$$4) f(x) = \cos(\sin \sqrt{2x+5})$$

$$f'(x) = -\sin(\sin \sqrt{2x+5}) \cos(\sqrt{2x+5}) \cdot \frac{1}{2\sqrt{2x+5}} \cdot 2$$

$$f'(x) = -\frac{\sin(\sin \sqrt{2x+5}) \cos \sqrt{2x+5}}{\sqrt{2x+5}}$$

$$f'(2) = -\frac{\sin(\sin \sqrt{2(2)+5}) \cos(\sqrt{2(2)+5})}{\sqrt{2(2)+5}}$$

$$f'(2) = -\frac{\sin(\sin(3)) \cos(3)}{3} \approx \underline{\underline{0.04647 \approx 0.05}}$$

$$5) f(x) = \tan(\cos \frac{x}{2})$$

$$f'(x) = \sec^2(\cos \frac{x}{2}) \cdot -\sin(\frac{x}{2}) \cdot \frac{1}{2}$$

$$f'(0) = \sec^2(\cos \frac{0}{2}) \cdot -\sin(\frac{0}{2}) \cdot \frac{1}{2} = \underline{\underline{0}}$$

$$6) 3y^2 + 6xy = 2x$$

$$6yy' + 6(y + y'x) = 2$$

$$6yy' + 6y + 6xy' = 2$$

$$6yy' + 6xy' = 2 - 6y$$

$$y' = \frac{2 - 6y}{6y + 6x}$$

$$y' = \frac{1 - 3y}{3y + 3x}$$

$$y'' = \frac{-3y'(3y+3x) - (1-3y)(3y'+3)}{(3y+3x)^2}$$

$$y'' = \frac{-9yy' - 9xy' - (3y'+3-9yy'-9y)}{(3y+3x)^2}$$

$$y'' = \frac{-9yy' - 9xy' - 3y' - 3 + 9yy' + 9y}{(3y+3x)^2}$$

$$y'' = \frac{-9xy' - 3y' - 3 + 9y}{(3y+3x)^2}$$

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$$y'' = \frac{-9x \left(\frac{7-3y}{3y+3x} \right) - 3 \left(\frac{7-3y}{3y+3x} \right) - 3 + 9y}{(3y+3x)^2}$$

$$y'' = \frac{-6x + 18xy - 7 + 9y^2}{(y+x)(3y+3x)^2} =$$

$$y''|_{(1,1)} = \frac{-6(1) + 18(1)(1) - 7 + 9(1)^2}{(1+1)(3(1)+3(1))^2} = \underline{\underline{0.27}}$$